

Detailed Project Scope and Specification

Vision-Only Leader-Follower Swarm Control on Constrained Edge Hardware

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Program: Master of Science in Electrical and Computer Engineering (ECE)

Credits: 4-Credit MS Project

Target Completion: First week of December 2026

1. Project Overview & Core Objectives

The primary objective of this project is to develop, simulate, and validate a decentralized, vision-based leader-follower autonomous robotic system. The system will eliminate the reliance on global mapping, centralized Wi-Fi fleet management, and computationally heavy or expensive LiDAR sensors. Instead, a follower Autonomous Mobile Robot (AMR) will maintain a precise geometric formation behind a leader AMR utilizing solely a localized RGB camera feed processed through a highly quantized neural network.

This thesis bridges advanced perception algorithms with classical control theory, proving that robust multi-agent coordination can be achieved on resource-constrained edge computing environments typical of modern, scalable robotics deployments.

2. System Architecture & Computational Graph

The software stack will be developed entirely within the ROS 2 Jazzy Jalisco middleware, utilizing isolated namespaces to simulate decentralized edge computation.

2.1 Perception Pipeline

- **Input:** Uncompressed RGB video stream (e.g., 640x480 at 30 FPS) from the follower robot's forward-facing camera plugin in Gazebo.
- **Inference Engine:** A YOLOv8-Nano object detection model. The model will be specifically trained/fine-tuned on synthetic data to recognize the unique rear-profile of the leader AMR.
- **Optimization:** To meet edge constraints, the weights will be quantized (INT8/FP16) and executed via TensorRT or ONNX Runtime to minimize latency and memory footprint.

2.2 State Estimation & Control Pipeline

- **Kinematic Extraction:** A ROS 2 node will subscribe to the bounding box data (x_center ,

y_center, width, height). The relative depth (distance) is estimated inversely proportional to the bounding box area, while the heading angle (yaw error) is derived from the horizontal offset of the bounding box center from the image frame's principal point.

- **Filtering:** An Extended Kalman Filter (EKF) will process these noisy geometric estimates, fusing them with the follower robot's internal wheel odometry to predict the leader's continuous state, effectively bridging gaps during temporary visual occlusions (e.g., dropped frames or motion blur).
- **Visual Servoing Controller:** A tuned PID control loop will translate the EKF's error signals into linear and angular velocity commands to maintain a designated target distance and heading.

[ROS 2 Computational Graph Concept]

/follower/camera/image_raw

|--> /yolov8_tensorrt_node (Outputs: vision_msgs/Detection2DArray)

|--> /relative_state_estimator (Outputs: geometry_msgs/PoseStamped)

|--> /ekf_smoothing_node

|--> /pid_visual_servo_controller

|--> /follower/cmd_vel (Twist commands to motors)

3. Hardware Constraints & Deployment Strategy

A critical aspect of this project's academic and industrial value is the deliberate limitation of computational resources, mimicking the realities of embedded hardware on commercial AMRs.

- **Host System Constraints:** Ubuntu 24.04 LTS natively running on an Intel Core i5 (12th Gen) with 16GB system RAM and an NVIDIA RTX 3050 Ti (4GB VRAM).
- **VRAM Management Protocol:** 4GB of VRAM presents a strict ceiling. Gazebo Harmonic's rendering engine will consume approximately 1.5GB to 2GB. The remaining VRAM strictly limits the neural network. Deploying unoptimized foundational models is impossible; thus, the TensorRT quantization step is a mandatory engineering requirement, forcing the YOLOv8-Nano footprint below 800MB.
- **CPU Multi-threading:** The 12th Gen i5's hybrid architecture will be explicitly managed using ROS 2 executor configurations (MultiThreadedExecutor) to ensure the EKF and PID control loops operate at a strict 50Hz frequency, independent of the variable inference latency of the perception node.

4. Simulation Environment & Testing Protocol

All validation will be conducted in Gazebo Harmonic, utilizing physics-accurate AMR models (e.g., TurtleBot4 or custom differential drive URDFs). The testing protocol will involve three distinct phases:

1. **Static Tracking:** Leader robot remains stationary; follower robot is spawned at various

angles and depths to verify initial bounding box and depth estimation accuracy.

2. **Dynamic Linear Tracking:** Leader robot executes predictable, straight-line trajectories with variable velocity profiles (acceleration/deceleration).
3. **Adversarial Trajectories:** Leader robot executes sharp turns, S-curves, and moves behind partial occlusions (e.g., a warehouse rack) to stress-test the EKF's predictive capabilities and the PID controller's recovery response.

5. Risk Assessment & Mitigation

Identified Risk	Impact Analysis	Mitigation Strategy
VRAM Exhaustion (OOM Error)	Gazebo and the ML model crash concurrently on the RTX 3050 Ti, halting simulation entirely.	Lower Gazebo rendering resolution/shadows. Enforce strict FP16 quantization on the ML model. Run Gazebo in headless mode if necessary for data collection.
Target Loss on Sharp Turns	The leader AMR moves out of the follower's 60-degree camera FOV, breaking the control loop.	Implement a "Search/Recovery" state machine. If bounding box confidence drops to 0, use the last known EKF velocity vector to pivot and re-acquire the target.
Schedule Overrun (December Deadline)	Failing to compile final thesis data in time for ECE departmental review.	Strictly limit the scope to simulation only. Exclude physical hardware deployment to eliminate hardware debugging variables.

6. Final Deliverables

- **Full ROS 2 Workspace (GitHub):** A well-documented repository containing the custom packages, launch files, and URDFs.

- **Simulation Benchmark Data:** CSV files mapping the tracking error (distance and yaw) over time during the adversarial trajectory tests.
- **Final Thesis Document:** A comprehensive technical paper detailing the methodology, mathematical formulations of the EKF/PID systems, and VRAM optimization techniques.
- **Video Demonstration:** High-quality screen recordings of the Gazebo environment showing the perception bounding boxes overlaid on the live follower camera feed, juxtaposed with the robots' physical movements.